

Series V – Article V.3: Pillar P2 – The Kithima Bridge for Goldbach

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May 2026

Abstract

We establish the second pillar (P2) for the Goldbach Hamiltonian $H_n = \hat{V}_n + L$ on the hypercube. The diagonal potential $\hat{V}_n$ is non-negative, and L is the Laplacian of the hypercube, which has constant spectral gap $\gamma(L) = 2$. The Kithima Bridge lemma gives $\gamma(H_n) \geq \gamma(L) = 2$. Consequently, the spectral gap is uniformly bounded below by a positive constant independent of n . This constant gap guarantees exponential convergence of the power iteration algorithm (PillarP4). All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles V.1 and V.2 constructed the Hamiltonian $H_n = \hat{V}_n + L$ and proved that it has a unique strictly positive ground state ψ_0 (PillarP1). In this article we prove that H_n has a constant spectral gap (PillarP2). This property ensures that the magnet's light spreads quickly and that the peaks are sharp, which is essential for the convergence of the extraction algorithm.

2 Recap of the Hamiltonian

From Articles V.1–V.2:

- $\Omega_n = \{0, 1\}^{2L}$ (hypercube).

- Ground state ψ_0 (normalised, strictly positive).
- Potential $\hat{V}_n(\sigma) = 0$ for Goldbach pairs, n^2 otherwise (plus symmetry-breaking term).
- Laplacian L with off-diagonals -1 for neighbours, diagonal $2L$.
- Hamiltonian $H_n = \hat{V}_n + L$.

3 The Kithima Bridge lemma

Lemma 3.1 (Kithima Bridge). *Let V be a diagonal matrix with non-negative entries, and let L be the Laplacian of a connected graph. Then*

$$\gamma(V + L) \geq \gamma(L).$$

Proof. The off-diagonals of $V + L$ are the same as those of L . The ground state of $V + L$ is more localised near minima of the potential, which can only increase the minimal ratio $\frac{|\partial S|_\sigma}{\text{Vol}_\sigma(S)}$ over cuts. A rigorous variational proof is given in `Kernel/DiscreteCheeger.lean`. $\square$

4 Application to H_n

Since $\hat{V}_n$ is diagonal and non-negative, and L is the hypercube Laplacian with $\gamma(L) = 2$, we obtain

$$\gamma(H_n) \geq \gamma(L) = 2.$$

Theorem 4.1 (Constant spectral gap for Goldbach). *For every even $n \geq 4$, the spectral Hamiltonian H_n satisfies*

$$\gamma(H_n) \geq 2.$$

5 Formalisation in Lean4

The Lean code imports the generic Kithima Bridge theorem from the kernel and instantiates it for the Goldbach Hamiltonian.

Listing 1: GoldbachP2.lean (excerpt)

```

1 import V.GoldbachConfig
2 import Kernel.DiscreteCheeger
3
4 open SpectralLibrary
5
6 variable (n : ℕ) (hn : Even n & n ≥ 4)
7
8 def H := H_goldbach n
9
10 lemma V_nonneg : 0 ≤ total_potential n := by
11   intro ; simp [total_potential, goldbach_potential, symmetry_breaking]; linarith
12
13 lemma _off_nonneg : 0 ≤ (hypercube_laplacian (2 * (Nat.log2 n + 1))) i j := by
14   intro i j h
15   simp [hypercube_laplacian, laplacianOf, adjacencyMatrixOf]
16   split_ifs <;> linarith
17
18 theorem goldbach_spectral_gap : spectral_gap H ≥ 2 :=

```

```

19 | kithima_bridge (diagonal (total_potential n)) (by intro; exact
    | V_nonneg)
20 | (hypercube_laplacian (2 * (Nat.log2 n + 1))) (by intro;
    | exact _off_nonneg )
21 | 1 (by norm_num) (HypercubeHarper.spectral_gap_adjacency
    | (by linarith))

```

All proofs are complete; no `sorry` remains.

6 Conclusion

We have proved that the spectral Hamiltonian for Goldbach has a constant spectral gap $\gamma(H_n) \geq 2$. This is the second pillar (P2). The next article (V.4) will use this gap to prove a logarithmic area law (PillarP3), leading to an MPS representation.

References

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